

FANUC Robot **series**

R-30*i*B AND R-30*i*B PLUS CONTROLLER

Pressure Control Function

USER'S GUIDE

MARUBFLPC06141E REV B

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1 OVERVIEW

1. OVERVIEW

New Body Structure materials require non-spot weld joining processes (mechanical fasteners, clinch, etc) Non-spot processes do not use weld controller, but are servo-driven and have similar pressure requirements. It can be difficult to support these applications with standard servogun option.

Flexible pressure control option makes it possible to support non-spot pressuring applications with servogun function. Flexible pressure support can also be used to add custom features to spot welding applications.

Basic guideline / procedure to implement flexible pressure control:

- prepare flowchart/control sequence for the application
- Define an instruction name
- Define subroutine names.
- Create tp programs for subroutines, based on flowchart / sequence
- Set instruction config parameters on config screen
- Define error recovery options
- Create tp programs for production part

Note:

- Clinch process is template / reference process for this user guide.
- Guideline presented on following pages specifically refers to clinch process.
- All settings / programs are equipment-specific.
- Equipment#1 is used throughout this guideline for simplicity.

2 INSTRUCTION DEFINITION

2. INSTRUCTION DEFINITION

Flexible pressure control option makes it possible to customize execution sequence for a servogun pressure instruction. Template provides structure and standard support. Application specific requirements are met by customizing the definition and creating tp programs.

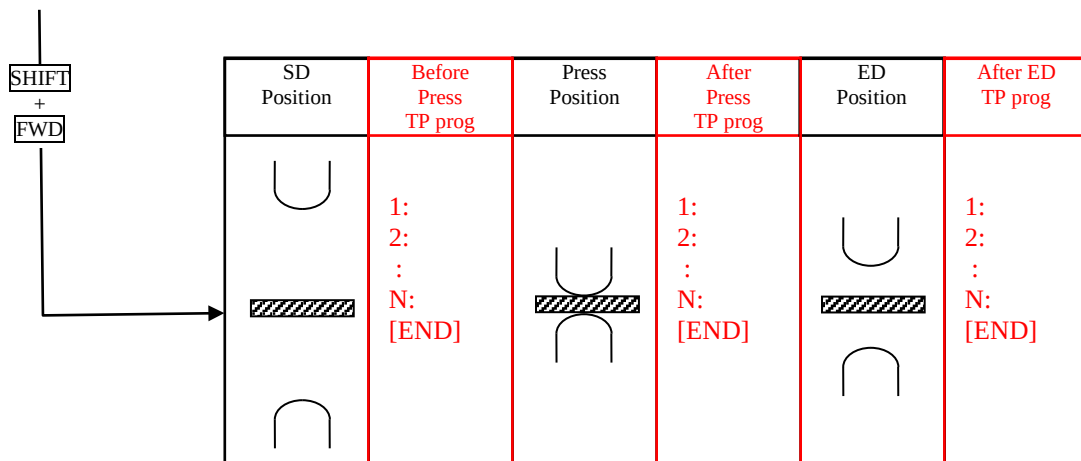
By default, flexible pressure instruction executes without subprogram calls (red boxes below). Enable and create subprograms as required for the application, as described in chapter.2.2

Default name for flexible pressure instruction is PRESS.

If a different name is required (for example CLINCH), then set it as described in chapter 3.

2.1 Instruction template

1: **PRESS*** [SD=, P=, t=, ED=]



2.2 Instruction definition procedure

Step 1:

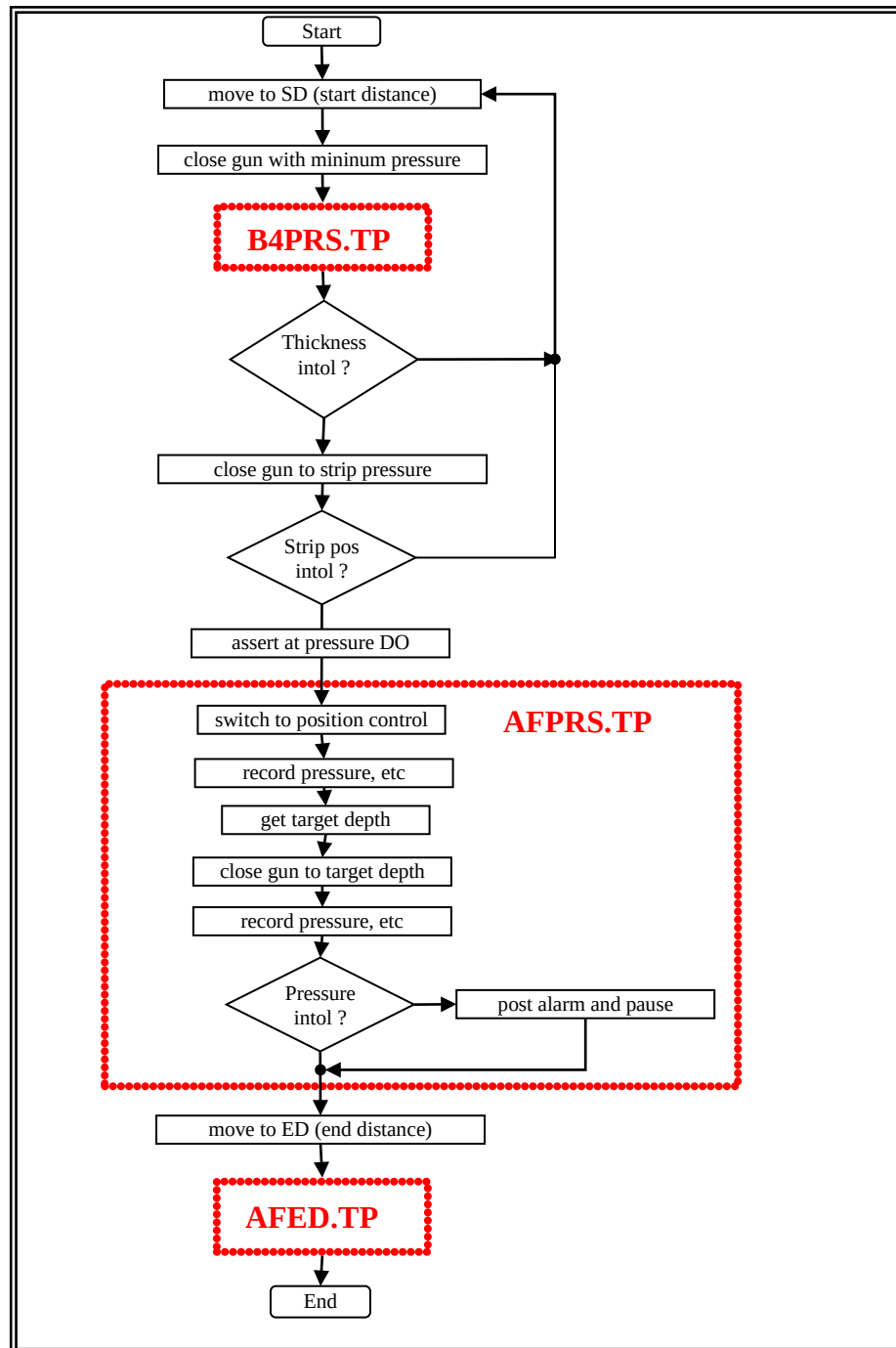
Create outline for application sequence before pressure and after pressure

Step 2:

Add outline info to boxes (b4prs.tp, afprs.tp, afopn.tp) in flowchart below.

Note:

- functionality outside of red boxes is provided by system, in flexible pressure control instruction.
- clinch process does not require customization at b4prs.tp, afed.tp, so boxes here are blank.



Step3:

Create tp subprogram(s) like afprs.tp from outlines in boxes above.

AFPRS.TP

- 1: CTRL_POSITION ;
- 2: CALL CLOSE_TO_DEPTH_EQ1(R[11])

Note1:

All tp instructions can used, EXCEPT for the following:

- SPOT
- PRESS_MOTN
- User press (“PRESS*” or name that was assigned to flexible pressure instruction)
- TIP_DRESS

Subprograms (like AFPRS.TP) cannot contain these instructions.

Note2:

If subprogram contains one of these instructions, the following alarm is posted when program executes:
SVGN-011 Cannot execute instruction(s).

Step4:

Configure instruction definition on user config screen. Refer to section 3

SETUP Servogun ^

USER PRESS: SET CONFIG / EQ:1 3/12

REQUESTED CFG: CLINCH*[SD, ^P^, t, ED]

STATUS: Apply needed <-

1 User press enable:	ENABLE	.
2 Instruction name:	CLINCH	.
3 Before press tp enable:	ENABLE	<-
4 Before press tpname:	_B4PRS	.
5 Execute weld schedule:	DISABLE	.
6 After press tp enable:	ENABLE	.
7 After press tpname:	AFPRS	.
8 Open after press:	ENABLE	.

Press F3 to apply change.

[TYPE] SHOW_CFG APPLY ENABLE DISABLE

Step5:

Add error recovery settings. Refer to section 4

SETUP Servogun ^

USER PRESS: RESUME CONFIG / EQ:1 1/5

Pause line	Resume line
*****	*****
SD motion line	Same line(resume)
Pre-Press motn ln ->	Same line(resume)
B4 press prog ->	Same line(resume)
Pressure line ->	Same line(resume)
AF press prog ->	Same line(resume)
ED motion line ->	Same line(resume)

[TYPE] [CHOICE]

Step6:

Select new instruction from page 3 of tp fkey instructions OR from PRESS_MOTN page of tp instruction list

A00 ^ i 1/1

[End]

CLINCH*

A00 ^ i 1/1

Pressure motion 1

1 Press_m[SD=,P=,t

2 CLINCH*[...]

3

4

5

6

7

8

Select item

[CHOICE]

Example**CLINCH_PROD.TP**

```

1: L P[1] 100% FINE
   : CLINCH*[SD=1,P=1,t=0.0,ED=1] ;
2: L P[2] 100% FINE
   : CLINCH*[SD=1,P=1,t=0.0,ED=1] ;
[END]

```

3 CONFIG SCREEN

3. CONFIG SCREEN

This screen is accessed from servogun general setup screen and configures control parameters for flexible pressure control (mnemonic instruction).

SETUP Servogun ^

GENERAL / EQ:1 Gun:1 15/15

6 Max Pressure(kgf): 500.0

Max Gun Torque(%): 55.0

7 Tip stick detect distance(mm): 5

8 Tip Wear Detection: < *DETAIL* >

9 Pressure Cal: COMP < *DETAIL* >

10 Tip Wear Standard: INCOMP < *DETAIL* >

11 Thickness check: < *DETAIL* >

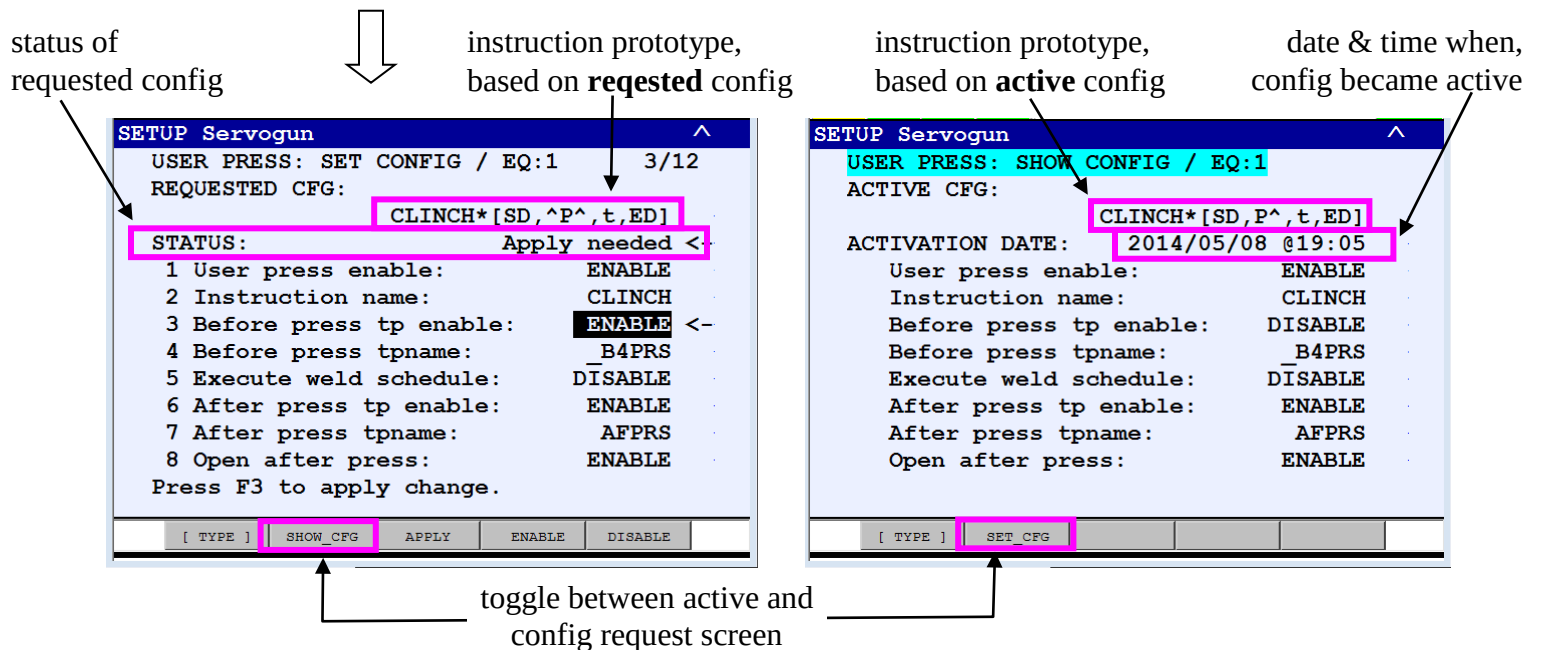
12 Gun Stroke limit: < *DETAIL* >

13 OverTorque protection: < *DETAIL* >

14 Gun Teach Setup: < *DETAIL* >

15 User pressure config: < *DETAIL* >

[TYPE] EQUIP GUN



Notes:

- Control parameters are specified on the SET CONFIG screen.
- Parameters active appear on SHOW CONFIG screen.
- Pressing F2 toggles between SET CONFIG screen and SHOW CONFIG screen.
- SHOW CONFIG screen is almost identical to set config screen, except:
 - title banner is blue
 - parameters cannot be modified.
 - activation date displays, instead of status
- the symbol <- appears to right of items on SET CONFIG screen that are NOT currently active (differ from SHOW config screen). Also, STATUS of requested config changes to APPLY NEEDED
- SET CONFIG screen parameters are applied at power-up, IF APPLY fkey is pressed before repower
- ^ symbol before / after P parameter identifies subprogram(s) that are enabled.

3.1 Pressure control config parameters

Item#	Item name	Description	Value
1	User press enable	Enable or disable execution of flexible pressure instruction. If instruction is disabled, SVGN-457 User press disabled is posted	DISABLE
2	Instruction name	Name to display for flexible pressure instruction. Maximum length of name = 21 chars. note: * is automatically appended to name to identify it as a custom instruction.	CLINCH
3	Before press tp enable	IF TRUE, tp program specified in Before pressure subroutine is called, and ^ appears to left of P symbol (in prototype)	ENABLE
4	Before press tpname*1	Routine name to call, before start of motion to pressure. IF null, REMARK is added to ~svg sequence IF not null, CALL is added to ~svg sequence Maximum length of name = 21 chars. Valid chars for name are alphabetic, numeric, and _ Name cannot start with a number. See appendix.	_B4PRS
5	Execute weld schedule	Execution weld control sequence after pressure is reached S= Appears in instruction if this option is enabled.	DISABLE
6	After press tp enable	IF TRUE, tp program specified in After press subroutine is called, and ^ appears to right of P symbol (in prototype)	ENABLE
7	After press tpname*1	Routine name to call, after pressure executes. IF null, REMARK is added to ~svg sequence IF not null, CALL is added to ~svg sequence Same name limitations as before pressure subroutine	_AFPRS
8	Open after press	Automatically move to open position, when AFTER_PRESS subroutine ends. If disabled, gun and robot remain at last position. ED= Appears in instruction, if this option is enabled.	DISABLE
9	After open tp enable	IF TRUE, tp program specified in After open tp name is called, and ^ appears to right of ED symbol (in prototype)	DISABLE
10	After open tp name	Routine name to call, after ED executes. IF null, REMARK is added to ~svg sequence IF not null, CALL is added to ~svg sequence Same name limitations as before pressure subroutine	_AFOPN
11	Number of user parameters	Number of extra integer parameters passed from instruction into subroutines. N1 = parameter 1, N2 = parameter 2 If number of parameters = 2, N1=,N2= are displayed in instruction.	0
12	Resume config	Action / operation at resume / continue	

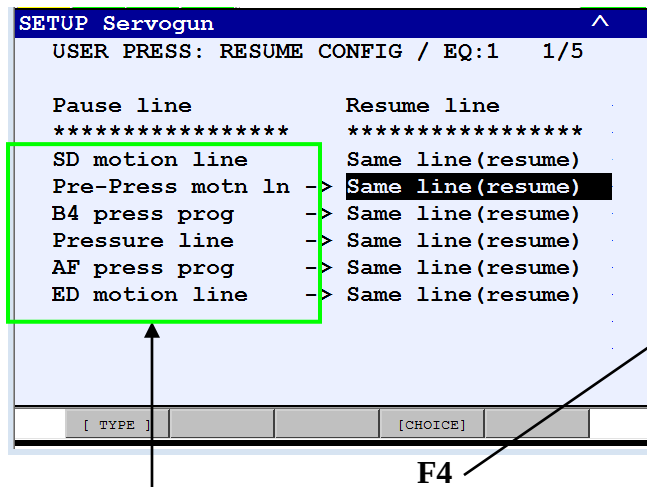
*1: equipment number in subroutine must match equipment number on config screen. I.E. each subroutines must have unique names.

3.2 Pressure control config fkeys

Fkey	Caption	Operation
F2	SHOW_CFG	Toggle between active and config request screen
F3	APPLY	Set apply flag. Requested config will be applied at power-up.
PREV		Return to general servogun setup screen

4 ERROR RECOVERY

4. ERROR RECOVERY



- 1 Approach motion
- 2 Press start motion
- 3 B4 press subprog
- 4 Apply pressure
- 5 AF press subprog
- 6 Open motion
- 7 AF open subprog

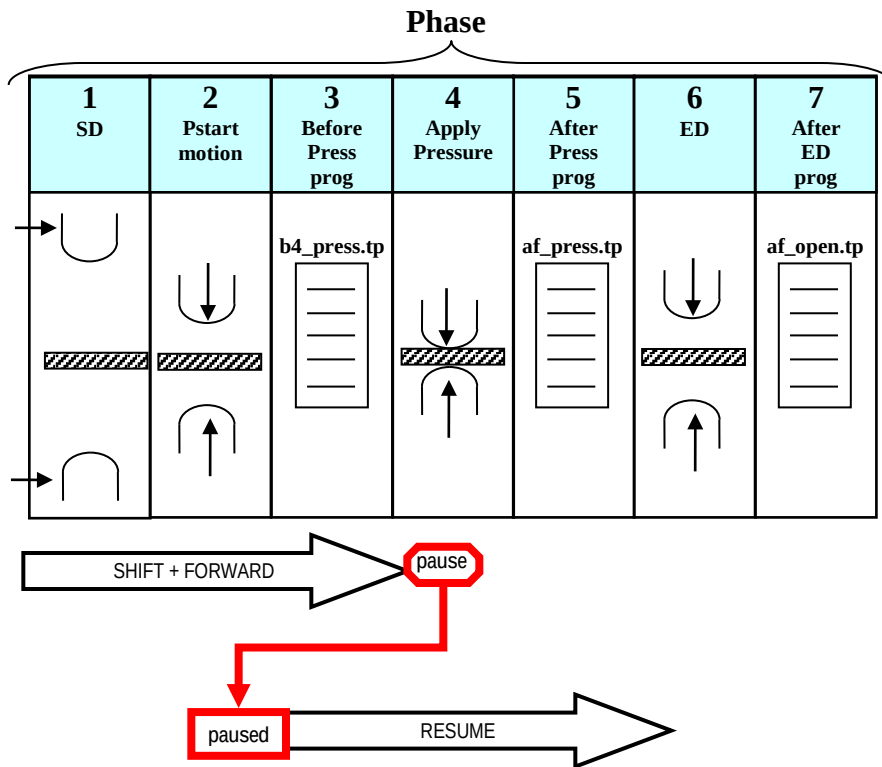
Defined phases list

Only defined (enabled) appear in this list. For example if ED is disabled then ED motion line does not appear in list

Resume line list

Same as defined phases list but two items are added:

- . same line (resume)
- . next line, main program



4.1 Error recovery resume options

As noted above, this screen defines the action that occurs at resume/continue, based on event that was active at time of pause. Again, option only appears in the list if that action is normal sequence contains that action .I.E. if open to ED = FALSE (on config screen), then Open motion will NOT appear in active pause list or resume list.

Move cursor to configure resume action for each pause action.

Resume option

Phase / line to resume	Description
Same line (resume)	Restart sequence at current line, when pause occurs at this action
Next line of main program	Restart sequence at next line of MAIN program, when pause occurs at this action
SD(approach motion)	Restart sequence at motion to SD, when pause occurs at this action
Press start motion	Restart sequence at motion to press start point, when pause occurs at this action
Before press subprog	Restart sequence at line 1 of b4prs subprog, when pause occurs at this action
apply pressure	Restart sequence at motion to pressure position, when pause occurs at this action
After press subprog	Restart sequence at line 1 of afprs subprog, when pause occurs at this action
ED (gun open) motion	Restart sequence at motion to ED, when pause occurs at this action
AF open subprog	Restart sequence at line 1 of afopen subprog, when pause occurs at this action

5 TP PROGRAM CREATION

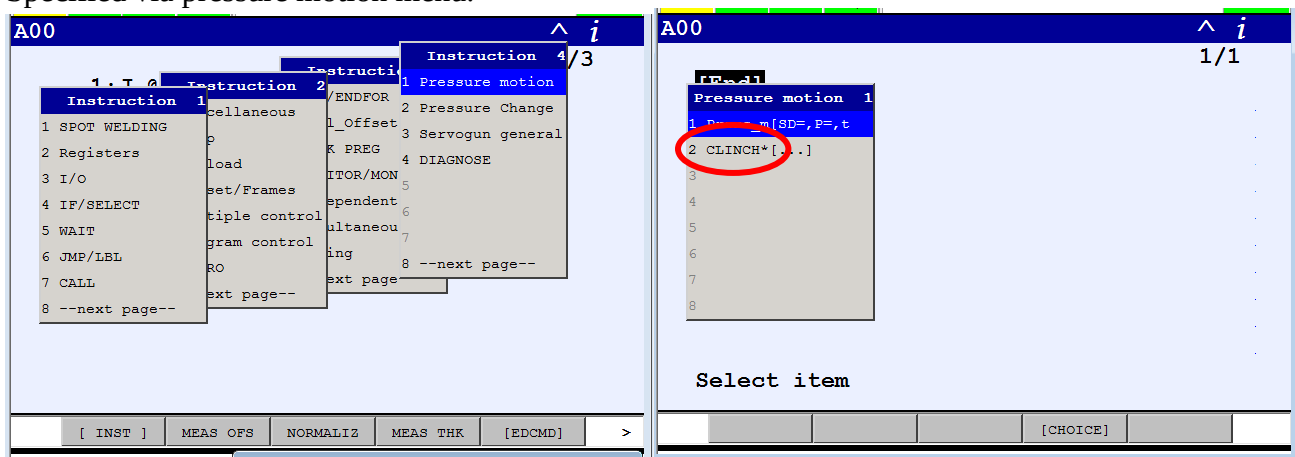
5. TP PROGRAM CREATION

User pressure instruction can execute with or without robot motion.

Standalone instructions might be desirable for pedestal application or application that does not require robot interaction.

5.1 Standalone flexible pressure instruction

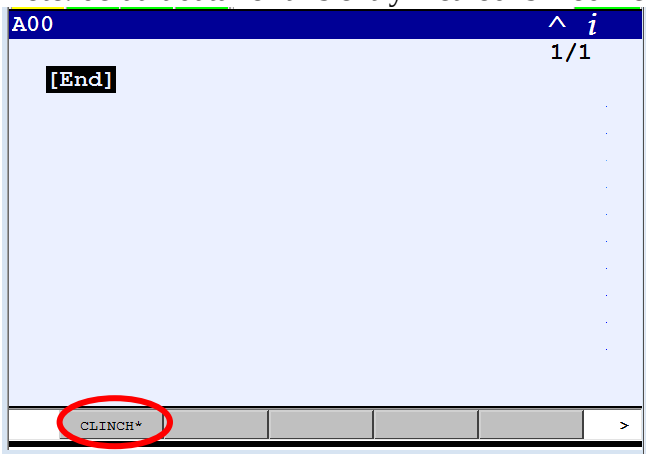
Specified via pressure motion menu.



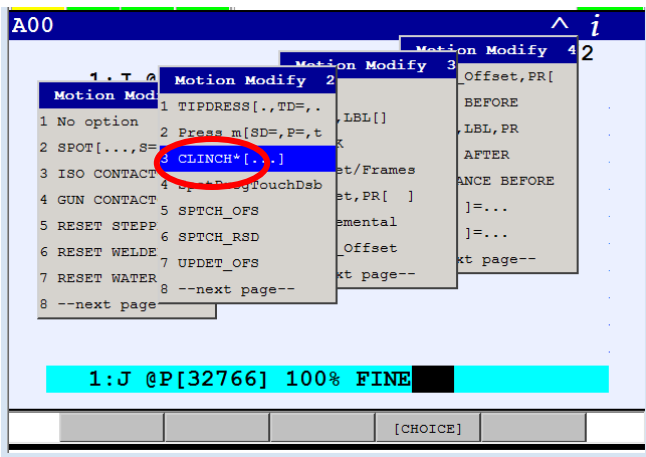
5.2 Motion option flexible pressure instruction

with motion can be specified by Fkey on page 3 of instruction menu

Note: default data for this entry method is modified by pressing Fkey without shift.



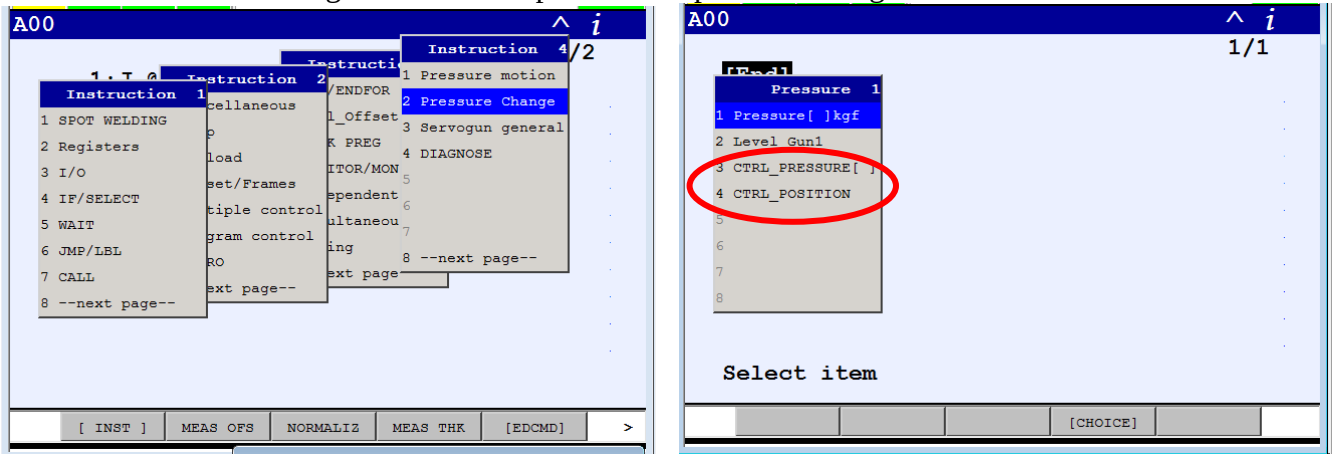
Flexible pressure instruction with motion can also be specified by appending instruction via motion modifier menu



5.3 Pressure control (switching) functions

Pressure control switching functions are used to change force control mode as required for application. In clinch case for example, gun closes with low pressure to find exact location of panel surface. Clinch application requires gun to close a pre-determined distance at this point. I.E. control mode must change from pressure to position control without moving the gun.

Pressure control switching functions are specified via pressure change menu



A APPENDIX

A. APPENDIX

A.1. Limitations

Available on v8.30 and higher.

Dual gun is not supported.

Maximum weld schedule value is 32766 (SHORT)

A.2. Memory

Function requires no additional memory.

TP programs that use flexible pressure instruction actually consume less memory than programs with spot instructions.

A.3. User press instruction details

CLINCH*[SD=1, P=1, t=0.0, s=1, ED=1, N1=1, N2=1] ;

Parameter	Description	Optional ?	min	max	Notes
SD	Start distance	no	1	32766	
P	Pressure schedule	no	1	256	
t	Thickness	no	0.0	999.9	
S	Weld schedule	yes	0	32766	Spot instruction supports long weld schedule (max =2147483646)
ED	End distance	yes	1	32766	
N1	Extra parameter	yes	1	32766	
N2	Extra parameter	yes	1	32766	

Note: all parameters support indirect value. I.E. SD=R[1] for example

A.4. Program name limitations

First character of name cannot be a number

Name cannot contain spaces

Name cannot contain the following characters

!	\	#	\$	%	&	\	(	)
+	-	.	/	:	;	<		
>	?	@	[	\\	]	^	`	

A.5. Variable Settings for CLINCH application

Variable	Setting	Notes
\$SGGUN1.\$SETUP.\$TC_ENB	TRUE	Enable thickness check before press
\$SGGUN1.\$SETUP.\$TC_ALM	TRUE	Check thickness at all points (note: if false, thickness is checked on request only)
\$SGGUN1.\$SETUP.\$TC_F2CMPR	TRUE	Check thickness at minimum pressure
\$SGGUN1.\$SETUP.\$TC_WSTPDLY	50	Settling time for 0kg thickness check
\$SGGUN1.\$SETUP.\$TC_MAX_UNDR	0.1	Max deviation from nominal thickness in negative direction
\$SGGUN1.\$SETUP.\$TC_MAX_OVR	0.1	Max deviation from nominal thickness in positive direction
\$SGSYSCFG.\$ALLOWAUTCNT	TRUE	Allow auto retry
\$SGCFG[1].\$THKERR.\$OPN_4TC_ALM	TRUE	Enable gun open for thickness check alarm
\$SGCFG[1].\$THKERR.\$RETLMTHK1LO	1	Retry thickness check 1 time, if undertol
\$SGCFG[1].\$THKERR.\$RETLMTHK1HI		Retry thickness check 1 time, if overtol
\$SGCFG[1].\$THKERR.\$THK1LOCNTCF	2	Action at modone = start ~svg per flg(s): do_2ndstep, and skip pen
\$SGCFG[1].\$THKERR.\$THK1HICNTCF	2	Action at modone = start ~svg per flg(s): do_2ndstep, and skip pen
\$SGCFG[1].\$THKERR.\$THK2LOCNTCF	2	Action at modone = start ~svg per flg(s): do_2ndstep, and skip pen
\$SGCFG[1].\$THKERR.\$THK2HICNTCF	2	Action at modone = start ~svg per flg(s): do_2ndstep, and skip pen
\$SGCFG[1].\$SHO_THK2	TRUE	Show thickness check after pressure
\$SGGUN1.\$THKCHK.\$THK2_CHKENB	TRUE	Enable thickness check at pressure
\$SGGUN1.\$THKCHK.\$THK2_CHKSEV	TRUE	Severity of thickness check at pressure fault = TRUE
\$SGGUN1.\$THKCHK.\$THK2_TOL_LO	0.1	Max difference from stripper thickness in negative direction
\$SGGUN1.\$THKCHK.\$THK2_TOL_HI	0.1	Max difference from stripper thickness in positive direction
\$SGGUN1.\$THKCHK.\$SETTL_TM	50	Settling time for thickness check at pressure
\$SGSCH1[1].\$WELD_PRESS	100	kgf setpoint for stripper pressure. If pressure is too low, then punch may be bad. If punch is too high ?
\$SGSCH1[1].\$PUSH_DEPTH	5	Button depth
\$SGGUN1.\$SETUP.\$CAL_SPEED1	140	Pressure speed
\$SGGUN1.\$SETUP.\$CAL_SPEED2	140	
\$SGDST1[1].\$DST_RBT	10.0m	SD & ED distance for robot
\$SGDST1[1].\$DST_GUN	10.0m	SD & ED distance for gun
\$SGCFG[1].\$CLR_CLSFLG2	0	Don't clear cls flag at AMR end for press_motn
\$SGUSRSYS.\$CFGAPLYDMSG	TRUE	Post message that config was applied for change notification

A.6. Program files for CLINCH application**AFPRS.TP**

```
1: CTRL_POSITION ;  
2: CALL CLOSE_TO_DEPTH_EQ1( R[11] )
```

CLINCH_PROD.TP

```
1: L P[1] 100% FINE  
   : CLINCH*[SD=1,P=1,t=0.0,ED=1] ;  
2: L P[2] 100% FINE  
   : CLINCH*[SD=1,P=1,t=0.0,ED=1] ;  
[END]
```

A.7. Alarm

Refer to the *Error Code Manual* for more information on the cause and remedy for each alarm.